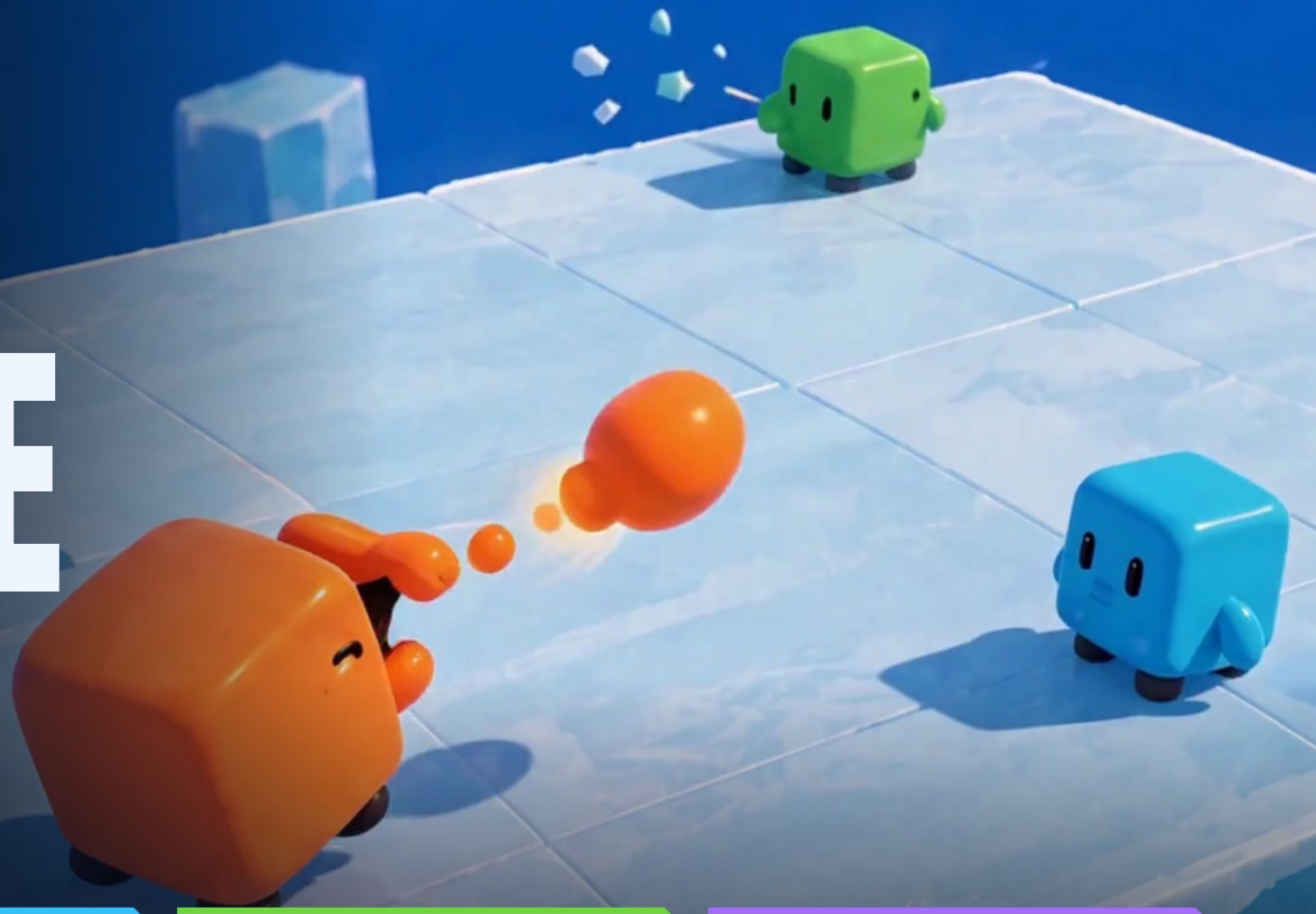
ICE RUMBLE



Concept render, AI-generated. Not gameplay.

WORKING TITLE

A LAN party game for 2–4 players: short physics mini-games, one shared scoreboard.

P1 Ezel Bayraktar
2220356197
READY

P2 Abdurrahim Özağaç
2220356033
READY

P3 Kürşat Hürkan Aydın
2230356008
READY

P4 Ardıl Bilal Ülkü
2230356051
READY

How it plays: one room, short rounds, one scoreboard

2–4 friends in one room, each at their own PC. One PC hosts; the others join over the local network. **Everyone plays the same cube, in one of four colours.**

01
LOBBY
Host starts a session. Others join with the host's local IP.

>

02
ROUND
Host picks the next mini-game. Everyone loads the same arena.

>

03
PLAY
Host runs the physics. Clients send input and draw the result.

>

04
PLACE
Round ends. Placement becomes points.

>

05
STANDINGS
Scoreboard updates. After the last round: overall winner.

NEXT MINI-GAME

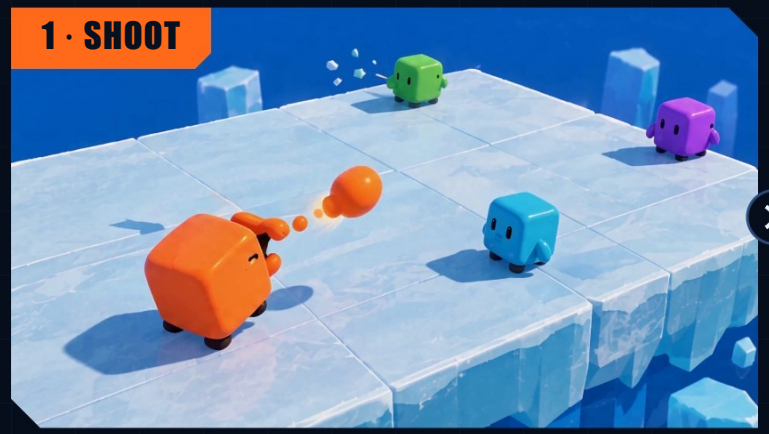
PLACEMENT POINTS · PROPOSED

1ST	■■■■■■■■■■■■■■■■■■■■	4 PTS
2ND	■■■■■■■■■■■■■■■■■■■■	3 PTS
3RD	■■■■■■■■■■■■■■■■■■■■	2 PTS
4TH	■■■■■■■■■■■■■■■■■■■■	1 PTS

BETWEEN ROUNDS

CARRIES OVER	Points. Nothing else accumulates.
CATCH-UP	The last round's winner earns slightly fewer points next round; last place earns slightly more.
RESETS	Mass and position, so one bad round does not snowball.

One value is your health, your ammunition and your weight



A cube spits out a piece of its own body. Every shot makes it smaller, hit or miss.

SHOOTER MASS

lost this exchange ▸ faded



The target loses mass on impact and is pushed back along the line of the shot.

TARGET MASS

lost this exchange ▸ faded



Less mass means the same hit pushes further. Off the edge is a knockout.

KNOCKBACK, SAME HIT

grows as mass drops

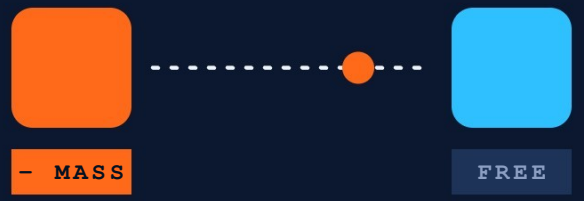
Shooting, getting hit and getting launched all read from one number the player can see: **the cube's size.**

Bars show direction only. Costs and knockback values are set in playtests.

Why both the shooter and the target lose mass

VARIANT A

Only the shooter pays



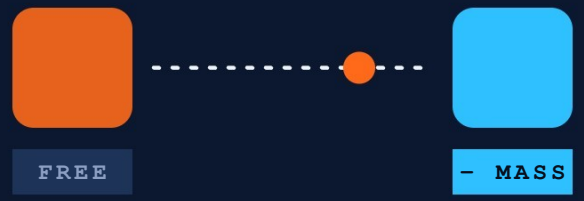
BEST PLAY

Shooting is pure loss. The best play is to never shoot and wait for the others to weaken each other.

× REWARDS HIDING

VARIANT B

Only the target pays



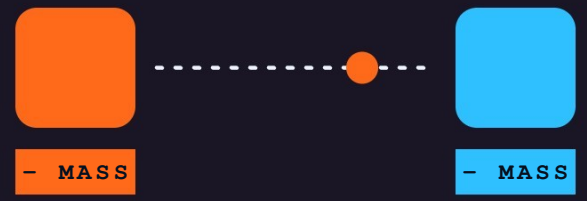
BEST PLAY

Shooting is free. The best play is to shoot nonstop in every direction.

× REWARDS SPAM

VARIANT C

Both pay: our rule



BEST PLAY

Every shot is a trade. A hit should take more from the target than the shot cost you, and a miss costs only you.

✓ REWARDS AIM AND TIMING

A shot only pays off when it lands, so **accuracy becomes the skill.** And never shooting scores nothing: points come from knockouts, not survival.

Exact costs are tuning values. We fix them in playtests, not on paper.



FIG 2 · Orange spits a piece of itself; Cyan is hit near the edge. Concept render, AI-generated. Not gameplay.

MINI-GAME 01

MASS KNOCKOUT

GOAL

Knock other cubes off the floating platform.

CONTROLS

Move, aim, spit. A short cooldown between shots.

SCORING

A knockout goes to the last cube that hit the one that fell. Most knockouts at the timer wins.

RESPAWN

After a short delay, at full mass.

ARENA

One flat square platform. Open edges, no walls.

MINI-GAME 02

ICE RACE

GOAL

Finish the laps first on a floating ice track.

CONTROLS

Steering changes acceleration, not velocity. Braking restores grip.

SKILL

Low sideways grip: carry speed through corners without sliding off.

FALLS

Respawn at the last checkpoint. You lose time, not the race.

PLACEMENT

Finish order. Unfinished players ranked by track progress.



FIG 3 · Racing a floating ice track; the HUD is part of the render. Concept render, AI-generated. Not gameplay.

One shared core; each mini-game plugs into it

MINI-GAMES

Mass Knockout
mass state · spit · knockback

Ice Race
ice profile · track · checkpoints

Next mini-game
new rules + one arena

MINI-GAME CONTRACT

`Setup(players) → Run() → Results() : placement`

SHARED GAMEPLAY

Character | Camera | Round state machine | Scoreboard | Lobby & UI

NETWORK

Host-authoritative LAN · Unity Netcode for GameObjects (candidate)

ENGINE

Unity · built-in rigidbody physics

WHY THIS SPLIT

Reuse

A new mini-game is new rules plus one arena. Character, camera, network and scores already exist.

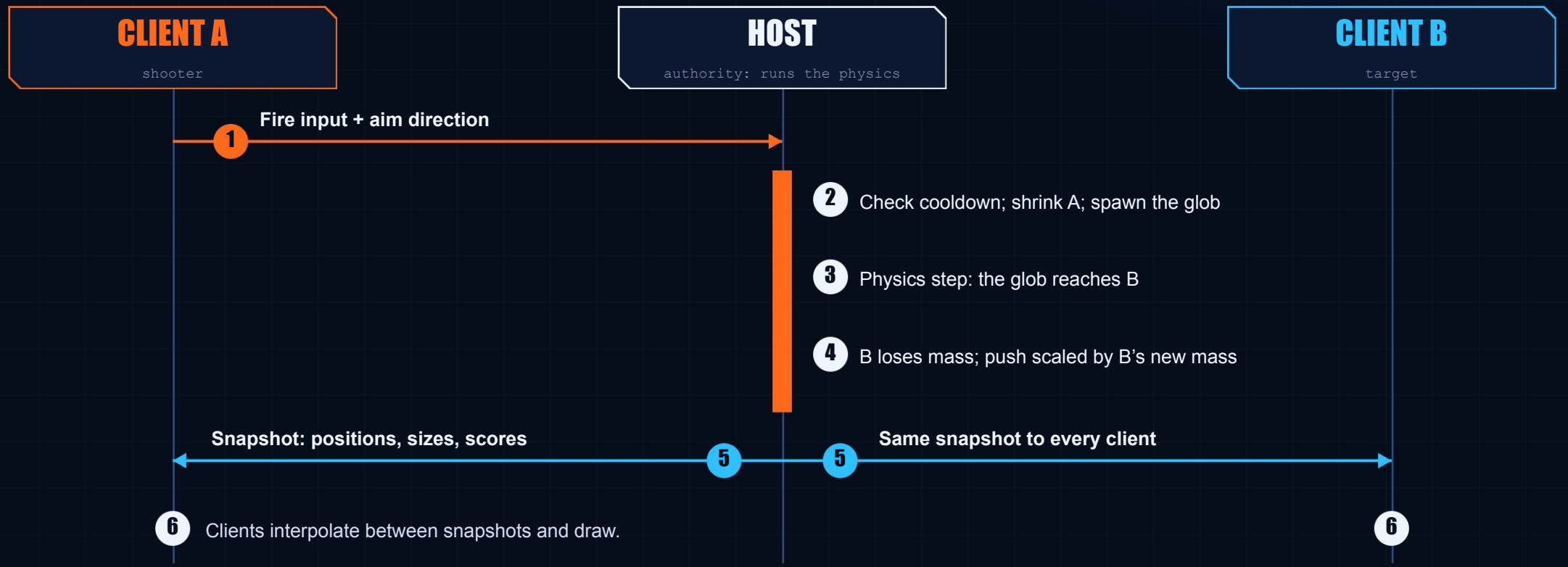
Per-game physics

Each mini-game sets its own profile: low grip on ice, firm grip in Knockout.

One truth

Everything that decides a result runs on the host, so every screen agrees.

One shot over the LAN, step by step



Your own movement is predicted locally, so it feels instant. Hits, mass changes and knockouts are decided only on the host. **Main risk:** keeping physics smooth on clients. LAN latency is low, and we tune the snapshot rate.

Social PC play is growing; LAN party games are rare

$43.6B

PC games revenue in 2025, up 12% year on year.

Newzoo [1]

$4.1B

Steam revenue of games with co-op in H1 2025, an all-time high (estimate).

Alinea [2]

70%

of US players aged 8+ play with other people.

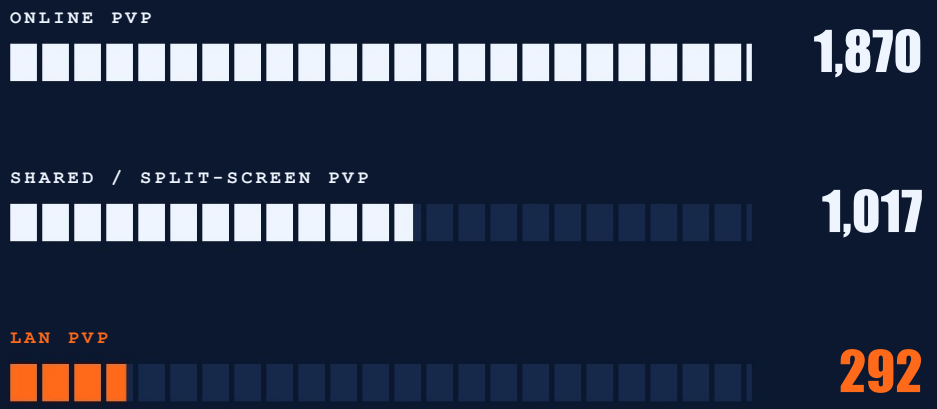
ESA 2026 [3]

≈8%

of ~3,490 Steam “Party Game” titles list LAN PvP.

Steam, Oct 2026 [4]

STEAM “PARTY GAME” TITLES BY MULTIPLAYER TAG



Steam store search, Oct 2026 [4]. Tags are user-applied.

RISK, STATED PLAINLY

An estimated 65.9% of 2025 Steam releases earned under $1,000 [5], while 5,863 games passed $100K in 2025 (Valve) [6]. **LAN is our differentiator, not a demand driver.**

Five games we are measured against



Party Animals

Recreate Games · 2023

VISUAL

Plush 3D animals, toy-like arenas.

MECHANIC

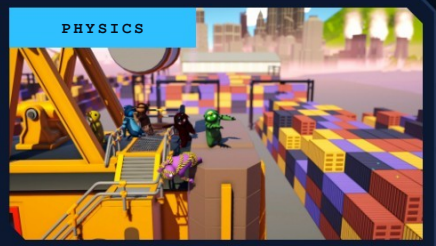
Physics brawling: grab, punch, throw rivals out.

MULTIPLAYER

Online and same-screen. No LAN listed.

FINANCIALS

Launch peak 104,174 players. Owner est. 2.5M–6.0M.



Gang Beasts

Boneloaf · 2017

VISUAL

Soft, gelatinous 3D people.

MECHANIC

Ragdoll brawling into stage hazards.

MULTIPLAYER

Same-screen and online. No LAN listed.

FINANCIALS

85% positive. Owner est. 4.6M–5.8M.



Fall Guys

Mediatonic / Epic · 2020

VISUAL

Candy-coloured beans, obstacle courses.

MECHANIC

Short race and survival rounds in a playlist.

MULTIPLAYER

Online, cross-platform only.

FINANCIALS

Free-to-play. Legacy Steam peak 172,213. Metacritic 80.



Pummel Party

Rebuilt Games · 2018

VISUAL

Colourful low-poly 3D, slapstick items.

MECHANIC

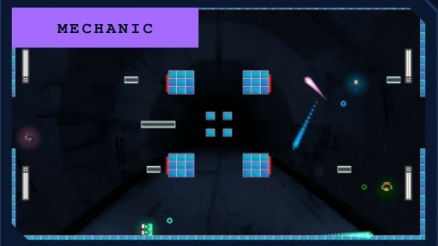
Board game with short competitive mini-games.

MULTIPLAYER

4–8 online and local. No LAN listed.

FINANCIALS

$14.99. Peak 20,957. Owner est. 1.5M –7.6M.



Squarena

Codeaclysm · 2022

VISUAL

Minimal flat 2D squares.

MECHANIC

Arena shooter: low health shrinks you; small bodies feel more recoil.

MULTIPLAYER

Up to 6 local. No LAN listed.

FINANCIALS

$4.99. Sales not disclosed.

Screenshots: official Steam media, © publishers. “Owner est.” = third-party estimates via SteamDB, not sales. 2026-10-17 [7].

Where we sit among them

	PARTY ANIMALS	GANG BEASTS	FALL GUYS	PUMMEL PARTY	SQUARENA	ICE RUMBLE
3D characters	●	●	●	●	○	●
Ranged attacks	◐	○	○	◐	●	●
Damage changes your body size	○	○	○	○	●	●
Less mass, further knockback	○	○	○	○	○	●
Several mini-games, one result	○	○	◐	●	○	●
Official LAN play	○	○	○	○	○	●
Plays with no internet	○	●	○	●	●	●

● Yes ◐ Partly ○ No ● Planned (Ice Rumble)

"Official LAN: No" = not advertised in the game's Steam features. "Partly": Party Animals and Pummel Party have some weapons or items; Fall Guys' playlist eliminates rather than scores. Party Animals needs internet even on one screen [8]. Ice Rumble column = planned design.

THE NEW LINK

No shipped game we found ties knockback to the mass you have lost.

Mass Knockout puts four links on one physical value the player can see. Links 01 to 03 have relatives in other games. **Link 04 we did not find.**



Prior-art search of shipped games, Oct 2026 [9-12]. Evidence on the next slide.

Closest prior art for each link, and what we did not find

LINK		CLOSEST PRIOR ART	STATUS
01	Spend your body	Osmos ejects its own matter to move, not to attack. Agar.io ejects mass.	DIFFERENT USE
02	Hits shrink you	Squarena (2D): lower health, smaller body. Morphies Law: hits move mass between robots.	PRECEDENT
03	Attack costs health	Crisol: Theater of Idols: your blood is your ammunition. Not physical mass.	PRECEDENT
04	Lighter, launched further	Smash Bros. raises knockback with a damage % counter, not mass. In Squarena, smaller bodies feel more of their own recoil.	NOT FOUND
01–04	All four in one 3D party game	No shipped game found.	NOT FOUND

Prior-art search of shipped games, Oct 2026 [9-12]. “Not found” means not found after searching.

Same rules, different builds: weight is also ammunition



FIG 4 · Heavy, balanced and light builds of the same character. Concept render, AI-generated.

Weight classes are standard in the genre (e.g. Mario Kart). The twist: here weight is also your ammunition, so a build changes how you play, not only how you look.

HEAVY

More starting mass: more shots to spend, harder to push. Moves slower.

MASS · AMMO

SPEED

BALANCED

The reference build every other one is tuned against.

MASS · AMMO

SPEED

LIGHT

Less starting mass: fewer shots, flies further when hit. Moves faster.

MASS · AMMO

SPEED

Framework for after the MVP. Pips are relative; stats are not designed yet.

Prototype: the mass rule simulated before any Unity code



WHAT IT IS

A deterministic physics simulation of the Mass Knockout rule, written in Three.js and rendered frame by frame.

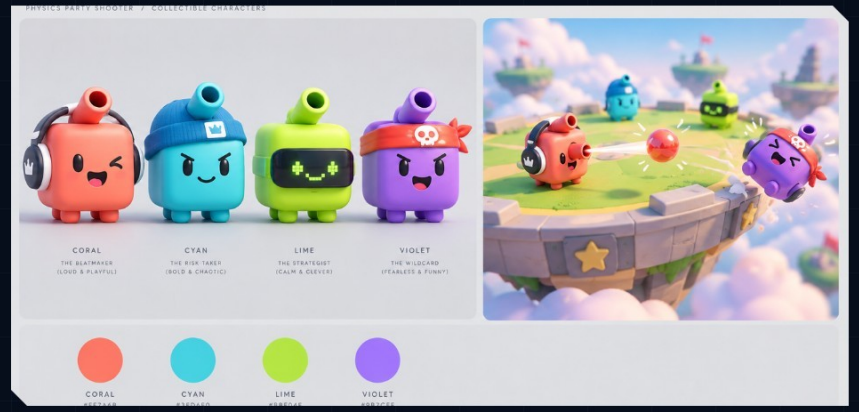
WHAT IT CHECKED

Over three shots the target shrinks and is pushed further each time. The escalating knockback reads clearly on screen.

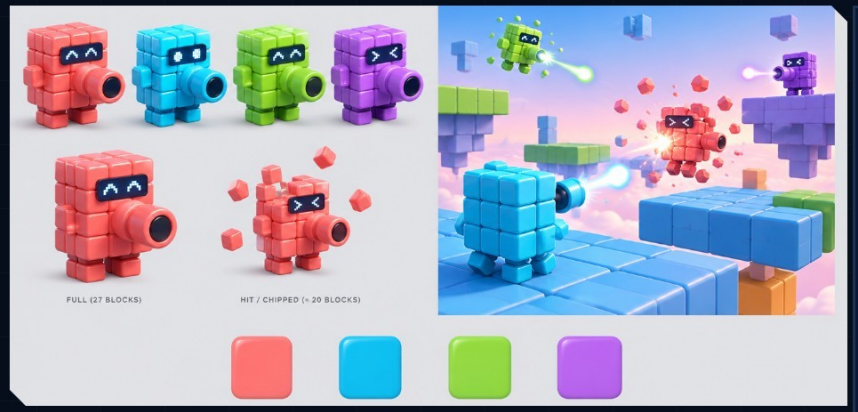
WHAT IT IS NOT

Not Unity, not networked, and not tuned. It tests whether the rule reads, not the final numbers.

Four art directions for the same cube



A VINYL TOY



B BUILDING BLOCKS



C TIN TOY



D JELLY

PURPOSE

Four looks for the same character, generated to choose a direction before modelling.

USED IN THIS DECK

The concept renders follow the glossy-toy direction.

PLAYER COLOURS

Orange	Cyan	Lime	Purple
#FF6A1A	#2EC0FF	#6CD63A	#A56BFF

All AI-generated drafts. None is gameplay.

Logo drafts



DRAFT A · WORDMARK

Title screen, deck cover, video end card.



DRAFT B · STACKED

Square formats: store capsule, lobby screen.



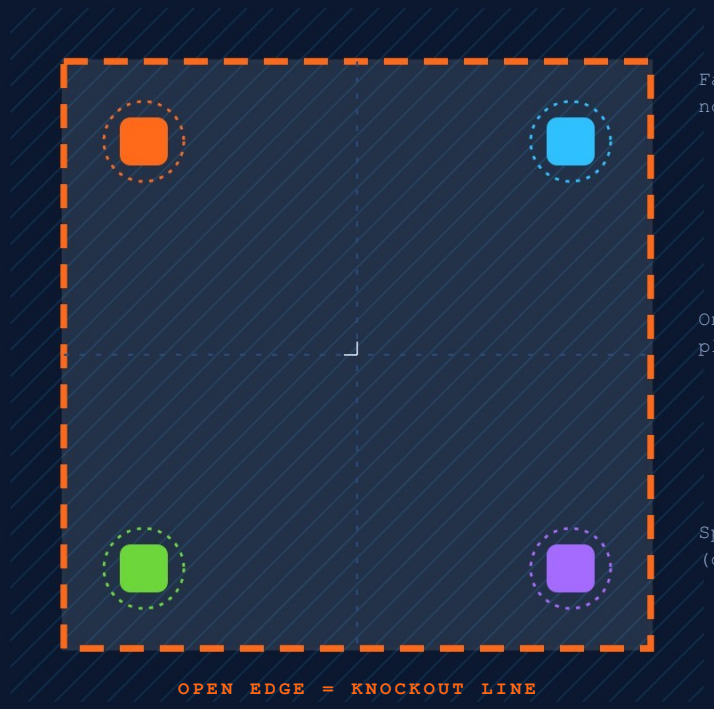
DRAFT C · ICON

Taskbar, build icon, favicon.

Drafts made for this proposal. The name is a working title; the logo is not final.

Level drafts, top-down: arena and track

MASS KNOCKOUT · ARENA DRAFT

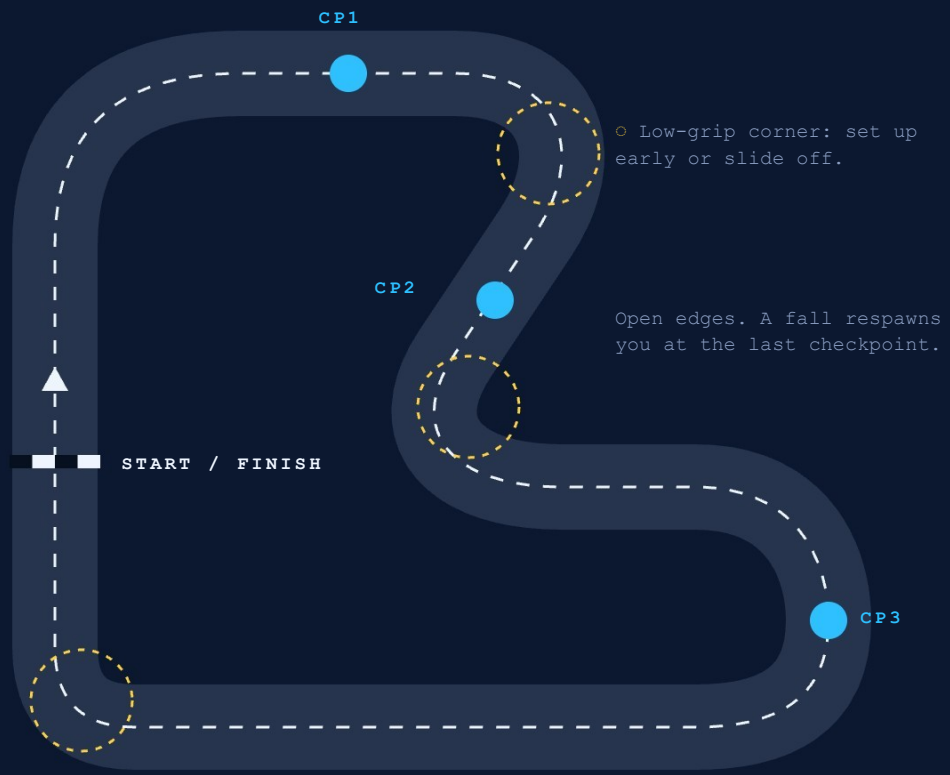


Fall zone. No walls, no cover.

One flat square platform.

Spawn near corners (draft).

ICE RACE · TRACK DRAFT



Low-grip corner: set up early or slide off.

Open edges. A fall respawns you at the last checkpoint.

Draft layouts for scope and blockout. Dimensions, spawn points and checkpoint count are not set.

Build order: shared core first, then each mini-game

Core	Character, camera, movement, fall detection
LAN	Host / join, host authority, sync
Ice Race	Ice movement, track, checkpoints, finish
Mass Knockout	Mass state, spit, hits, knockback, respawn
Session	Lobby, rotation, scoreboard, catch-up scoring
Tune & polish	Playtests; trails, hit flash, UI



ORDER IS FIXED · DATES AND OWNERS: TBD

RISK 01
NETWORKED PHYSICS

Clients disagree on a knockback.

Host simulates; clients only render and interpolate.

RISK 02
BALANCE

Stalling, or a leader snowballing.

Score knockouts; full-mass respawn; timer; catch-up scoring.

RISK 03
SCOPE

Too many mini-games for one term.

Exactly two in the MVP. More only once both are fun.

FINAL STANDINGS

ICE RUMBLE

1ST 2–4 players, one LAN, no internet.

2ND Two mini-games on one shared Unity core.

3RD Every shot costs the shooter mass, so aim matters more than volume.

QUESTIONS?

Ezel Bayraktar · Abdurrahim Özağaç · Kürşat Hürkan Aydın · Ardıl Bilal Ülkü



Sources

[1] Newzoo final 2025 figures, via GameDev Reports, 2026-06-26. gamedevreports.substack.com/p/newzoo-gaming-market-surpassed-200

[2] Alinea Analytics, games with co-op on Steam, H1 2025 (proprietary estimates). alineaanalytics.substack.com/p/games-with-co-op-generated-over-4

[3] ESA Essential Facts 2026 (survey Feb 2026, n=13,545). theesa.com

[4] Steam store search, tag “Party Game” × LAN PvP / Online PvP / Shared-Split-screen PvP, 2026-10-17. Tags are user-applied.

[5] Game Developers, 2025 Steam revenue distribution (Gamalytic estimates), 2026-08-21. game-developers.org/2025-steam-game-revenue-distribution

[6] PC Gamer, Valve at GDC 2026: 5,863 games over $100K on Steam in 2025. pcgamer.com

[7] Steam store pages and SteamDB charts, apps 1260320, 285900, 1097150, 880940, 1858400: prices, peaks, reviews, owner estimates. 2026-10-17.

[8] GameRevolution / Prima Games: Party Animals requires an internet connection, including split-screen.

[9] Osmos official site, osmos-game.com. [10] Squarena on Steam (app 1858400); codeaclysmgames.com/squarena-details.

[11] Morphies Law official site, morphieslaw.com/game. [12] Crisol: Theater of Idols, Blumhouse Games and Steam (app 1790930).

Images: all game visuals are AI concept renders (GPT Image via Higgsfield; MiniMax H3) from our own reference art, except competitor screenshots (official Steam media). Logo and level drafts made for this proposal.